

Mark Vincent Francisco

CS-499: Computer Science Capstone 2026

Professor Joseph Conlan

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Enhancement Two: Algorithms and Data Structure

Narrative:

1. Briefly describe the artifact. What is it? When was it created?

The selected artifact is an Event Planning mobile Android application that was initially created in my CS-360: Mobile Architect & Programming course back in March, 2026. This application was designed to assist users in storing upcoming events within a database along with pertinent information such as the title and description of the event, as well as the date of the event and respective phone number to send an SMS if an event is within 7 days of the current day. The application also implements a user interface for entering user login credentials to access the event UI or to create a new account and store it in the database. Once the user selects the login button, the user is able to access the user interface that allows them to create, delete, and update an event. In last week's enhancements, a login button has been applied along with input validations for username, password, date, and phone numbers to ensure that secure coding practices are being applied.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item?

What specific components of the artifact showcase your skills and abilities in algorithms and data structure? How was the artifact improved?

The reason why I selected this artifact for inclusion in my ePortfolio is because this particular artifact is able to demonstrate all course outcomes in one codebase as well as satisfy the required enhancements per categories A, B, and C. This artifact is an excellent piece to include with my GitHub repository because it demonstrates to potential employers fundamental computer science concepts that I have learned throughout my journey in college.

The specific component of this artifact that showcases my skills and abilities in the algorithm and data structure is within the `checkUpcomingEvents()` function of the `EventController.java` file. The standalone `ArrayList` for 'events' was replaced with a min heap (priority queue) data structure, which is a type of binary tree that helps with storing events in chronological order based on the date entered by the user. Essentially, the earliest date is stored at the root of the min heap which optimizes iteration of the list of events by breaking out of the loop once reaching an event that is greater than 7 days. This data structure is much better because the program avoids having to iterate through each and everyone of the events stored within the `ArrayList` which makes it incredibly inefficient if the application were to be operating at scale.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

Yes, I believe that I have closely met one of course outcomes that I had planned to meet with this enhancement which was:

- a. Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution, while managing the trade-offs involved in design choices (data structures and algorithms)

Thus far, I do not have any updates to my outcome-coverage plans and that my current plans are on track to fully fulfilling the course outcome requirements.

4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

What I learned as I was improving upon the Android mobile application artifact was how important it is to understand data structures exceptionally well, especially when it comes to handling user data at scale because system crashes are very prominent if not optimally implemented. I had to research several data structures that would best fit the checkupcomings() event use case in order to conservatively improve a system. The challenges that I faced while making improvements to this function in fulfillment of Category B was attempting to understand how min heaps work and how it translates into code written in Java. Also, figuring out a process to parse the date, and putting it into Calendar format by using Java's Calendar library was particularly confusing for me.